

AgisoftRep_K4_11522005_Syifa Syakira

Hasil Pengolahan Citra Drone

24 October 2024



Survey Data

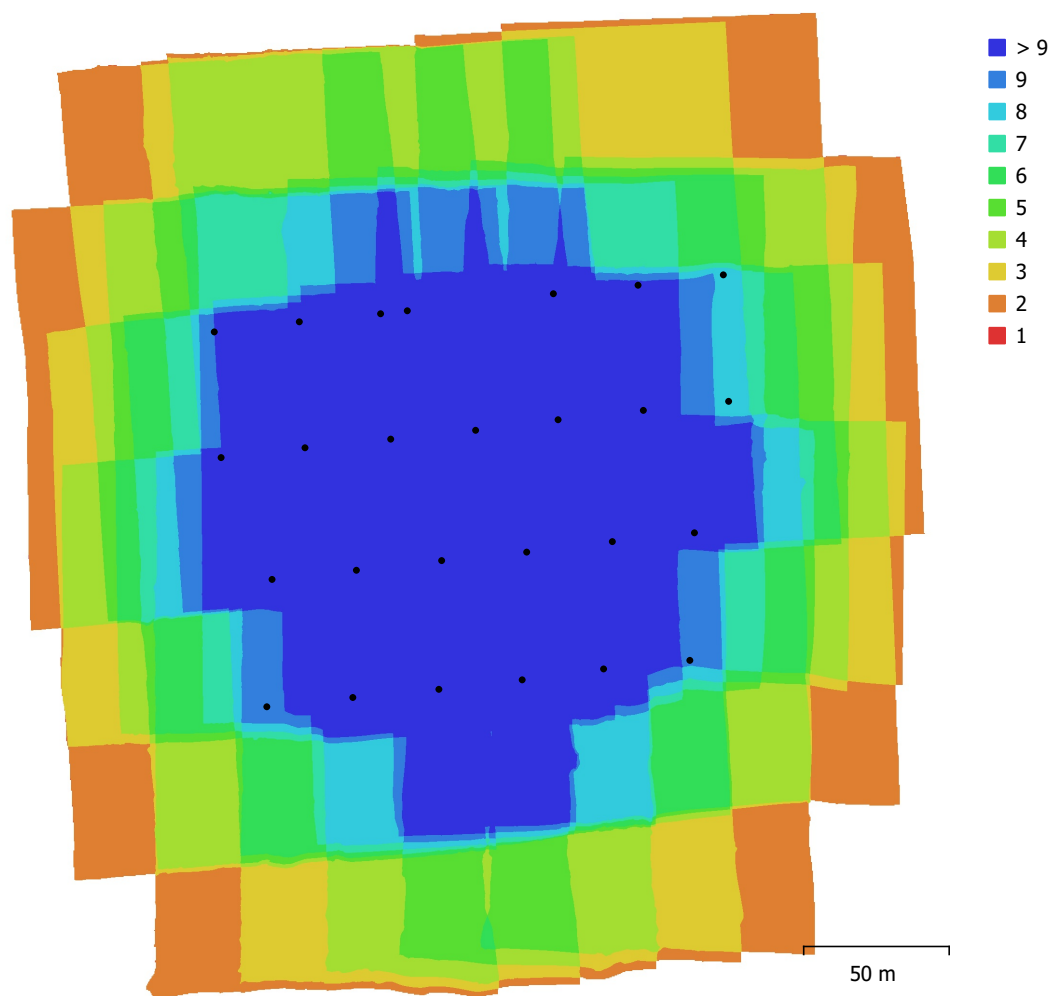


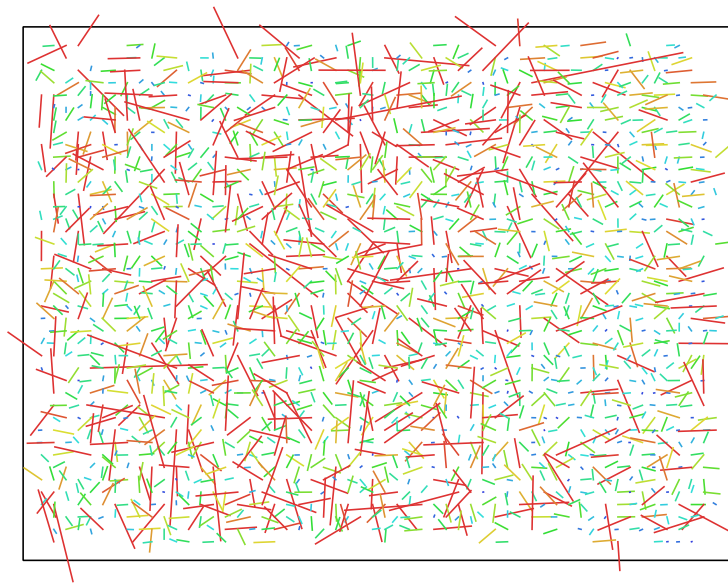
Fig. 1. Camera locations and image overlap.

Number of images:	26	Camera stations:	26
Flying altitude:	138 m	Tie points:	2,157
Ground resolution:	3.83 cm/pix	Projections:	6,523
Coverage area:	0.0915 km²	Reprojection error:	6.05 pix

Camera Model	Resolution	Focal Length	Pixel Size	Precalibrated
FC6310 (8.8mm)	4864 x 3648	8.8 mm	2.61 x 2.61 µm	No

Table 1. Cameras.

Camera Calibration



8 pix

Fig. 2. Image residuals for FC6310 (8.8mm).

FC6310 (8.8mm)

26 images

Type
Frame

Resolution
4864 x 3648

Focal Length
8.8 mm

Pixel Size
2.61 x 2.61 μm

	Value	Error	F	Cx	Cy	K1	K2	K3	P1	P2
F	3322.67	2.5e+02	1.00	0.73	-0.78	0.10	-0.59	0.75	0.36	0.35
Cx	3.47249	7.9		1.00	-0.53	0.07	-0.43	0.54	0.57	0.30
Cy	46.988	5.8			1.00	0.03	0.46	-0.58	-0.23	-0.02
K1	0.00353975	0.011				1.00	-0.47	0.41	0.41	-0.19
K2	-0.031189	0.016					1.00	-0.97	-0.27	-0.16
K3	0.0299244	0.018						1.00	0.32	0.23
P1	0.00154736	0.00035							1.00	0.05
P2	0.00120183	0.00025								1.00

Table 2. Calibration coefficients and correlation matrix.

Camera Locations

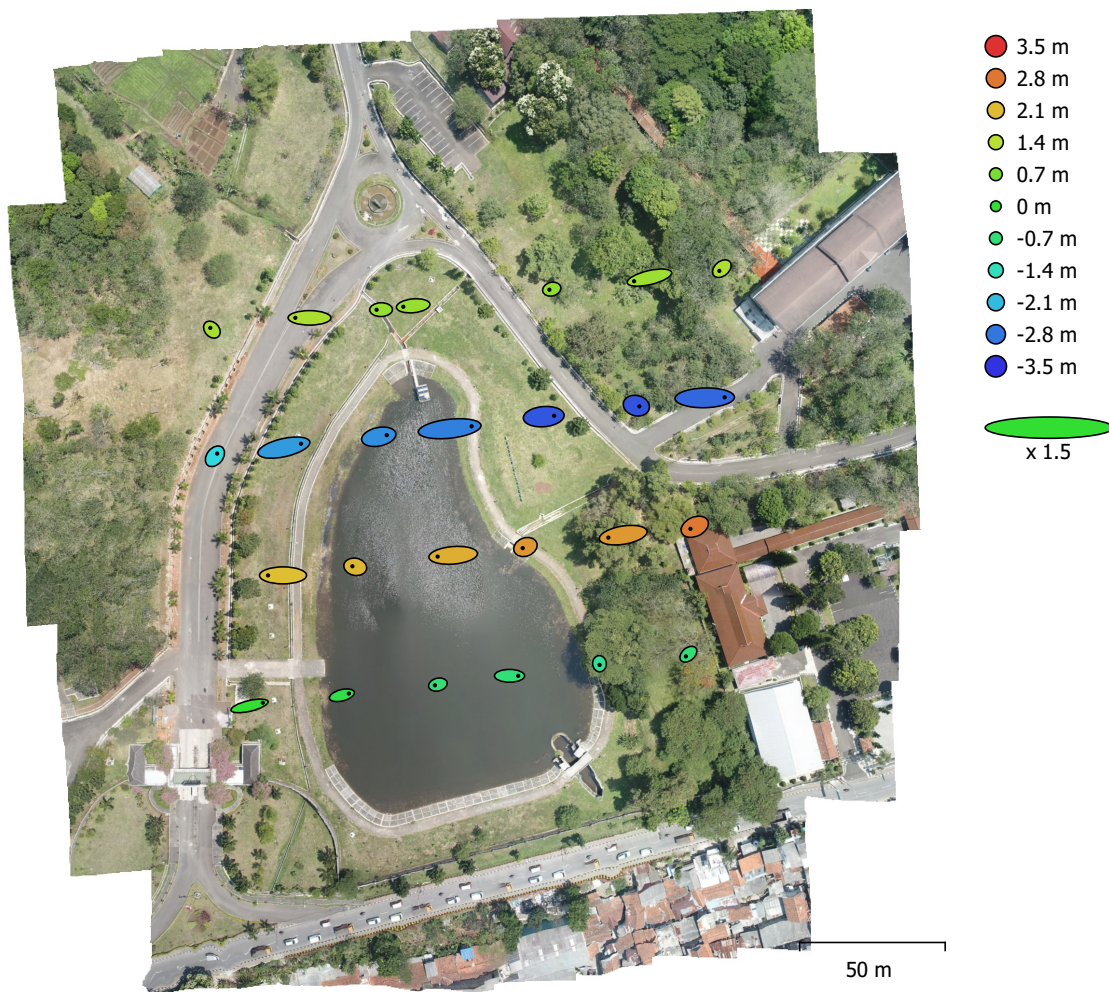


Fig. 3. Camera locations and error estimates.

Z error is represented by ellipse color. X,Y errors are represented by ellipse shape.
Estimated camera locations are marked with a black dot.

X error (m)	Y error (m)	Z error (m)	XY error (m)	Total error (m)
4.88835	0.840685	1.92876	4.96011	5.32192

Table 3. Average camera location error.
X - Longitude, Y - Latitude, Z - Altitude.

Digital Elevation Model

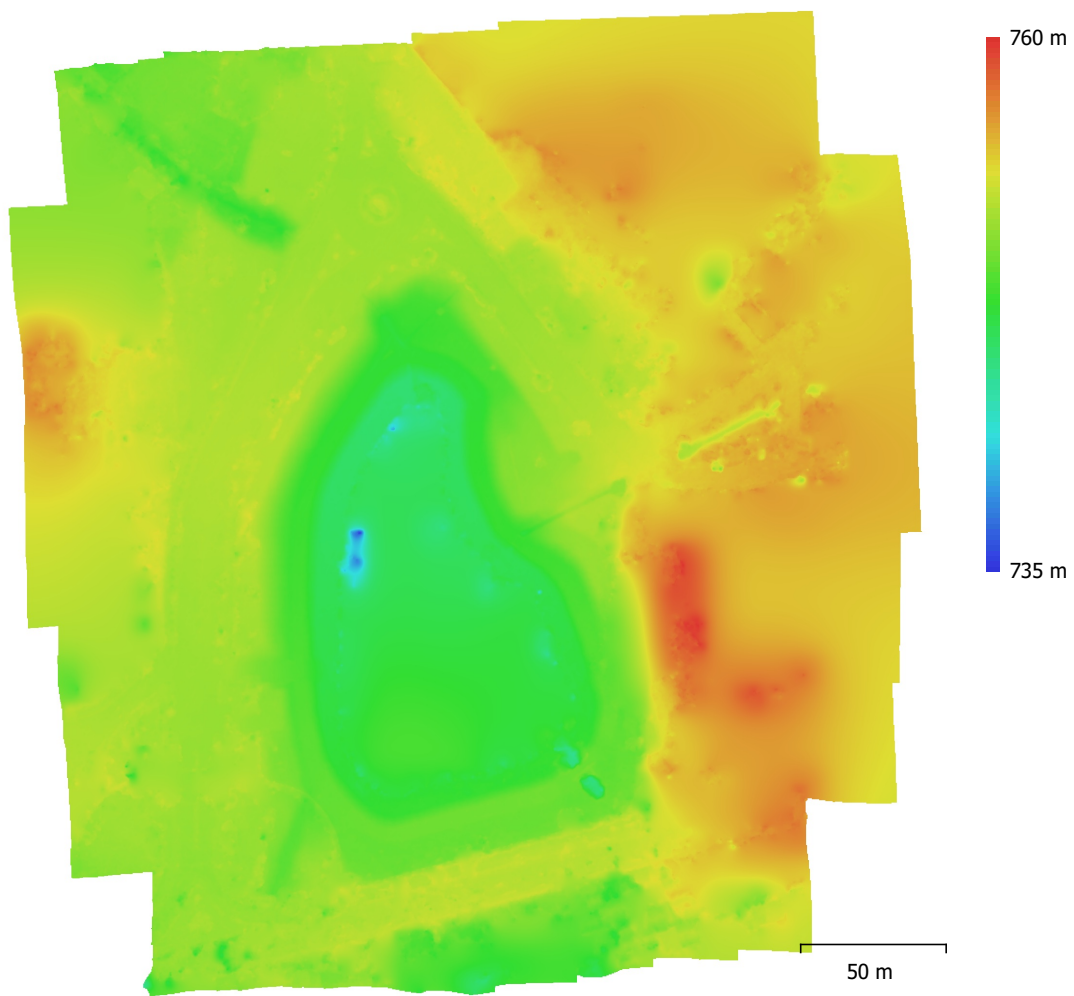


Fig. 4. Reconstructed digital elevation model.

Resolution: 30.6 cm/pix
Point density: 10.7 points/m²

Processing Parameters

General

Cameras	26
Aligned cameras	26
Coordinate system	WGS 84 (EPSG::4326)
Rotation angles	Yaw, Pitch, Roll

Point Cloud

Points	2,157 of 2,542
RMS reprojection error	0.195258 (6.04943 pix)
Max reprojection error	0.596553 (40.2787 pix)
Mean key point size	31.3621 pix
Point colors	3 bands, uint8
Key points	No
Average tie point multiplicity	3.25295

Alignment parameters

Accuracy	Lowest
Generic preselection	Yes
Reference preselection	Source
Key point limit	40,000
Key point limit per Mpx	1,000
Tie point limit	4,000
Exclude stationary tie points	Yes
Guided image matching	No
Adaptive camera model fitting	No
Matching time	44 seconds
Matching memory usage	263.34 MB
Alignment time	2 seconds
Alignment memory usage	19.11 MB
Date created	2024:10:24 01:34:14
Software version	1.7.4.12511
File size	190.66 KB

Depth Maps

Count	26
Depth maps generation parameters	
Quality	Low
Filtering mode	Aggressive
Processing time	32 seconds
File size	8.45 MB

Dense Point Cloud

Points	972,168
Point colors	3 bands, uint8

Depth maps generation parameters

Quality	Low
Filtering mode	Aggressive
Processing time	32 seconds

Dense cloud generation parameters

Processing time	15 seconds
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Ground points classification parameters

Max angle (°)	15
Max distance (m)	1
Cell size (m)	50

Classification time	2 seconds
Classification memory usage	61.75 MB
Date created	2024:10:24 01:36:20
Software version	1.7.4.12511
File size	13.99 MB
Model	
Faces	19,690
Vertices	10,181
Vertex colors	3 bands, uint8
Texture	4,096 x 4,096, 4 bands, uint8
Depth maps generation parameters	
Quality	Low
Filtering mode	Aggressive
Processing time	32 seconds
Reconstruction parameters	
Surface type	Height field
Source data	Dense cloud
Interpolation	Enabled
Strict volumetric masks	No
Processing time	1 seconds
Memory usage	35.30 MB
Texturing parameters	
Mapping mode	Generic
Blending mode	Mosaic
Texture size	4,096
Enable hole filling	Yes
Enable ghosting filter	Yes
UV mapping time	1 minutes 38 seconds
UV mapping memory usage	305.71 MB
Blending time	1 minutes 23 seconds
Blending memory usage	1.81 GB
Blending GPU memory usage	1.03 GB
Date created	2024:10:24 02:11:22
Software version	1.7.4.12511
File size	27.33 MB
Tiled Model	
Texture	3 bands, uint8
Depth maps generation parameters	
Quality	Low
Filtering mode	Aggressive
Processing time	32 seconds
Reconstruction parameters	
Source data	Dense cloud
Tile size	256
Face count	Low
Enable ghosting filter	No
Processing time	4 minutes 25 seconds
Memory usage	3.00 GB
Date created	2024:10:24 02:34:15
Software version	1.7.4.12511
File size	72.19 MB
DEM	
Size	1,229 x 1,438
Coordinate system	WGS 84 / UTM zone 48S (EPSG::32748)
Reconstruction parameters	
Source data	Dense cloud

Interpolation	Enabled
Processing time	2 seconds
Memory usage	35.52 MB
Date created	2024:10:24 02:39:31
Software version	1.7.4.12511
File size	3.75 MB
Orthomosaic	
Size	8,183 x 8,815
Coordinate system	WGS 84 / UTM zone 48S (EPSG::32748)
Colors	3 bands, uint8
Reconstruction parameters	
Blending mode	Mosaic
Surface	DEM
Enable hole filling	Yes
Enable ghosting filter	No
Processing time	50 seconds
Memory usage	1.28 GB
Date created	2024:10:24 02:41:22
Software version	1.7.4.12511
File size	569.32 MB
System	
Software name	Agisoft Metashape Professional
Software version	1.7.4 build 12511
OS	Windows 64 bit
RAM	7.70 GB
CPU	11th Gen Intel(R) Core(TM) i5-1135G7 @ 2.40GHz
GPU(s)	NVIDIA GeForce MX330